



The power of words in online auctions: how quality rhetoric and vagueness in item listings influence herding bias and wordiness effects

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Abstract

This study investigates how listing characteristics influence purchasing decisions in online auctions. Specifically, it examines whether an item description's length (i.e., wordiness) and herding bias affect the likelihood of a successful sale and explores the moderating effects of quality rhetoric and linguistic vagueness. We quantify these rhetorical characteristics using computer-aided text analysis on 230,270 auction descriptions. Our findings confirm that herding bias positively predicts sales probability, while an optimal listing length exists beyond which excessive verbosity diminishes sales likelihood. Quality rhetoric and vagueness negatively moderate the herding effect, which suggests that overt attempts to establish legitimacy under information asymmetry may generate skepticism. Furthermore, quality rhetoric amplifies the negative effect of excessive wordiness on sales, whereas vague terminology is positively associated with improved sales outcomes in wordier descriptions. These findings highlight the nuanced interplay between textual cues and consumer behavior in digital marketplaces, offering guidance for optimizing auction listing strategies.

Keywords Online auctions · Web data · Herding bias · Rhetoric · Wordiness · Vagueness

Abbreviations

Buyer premium	An additional service fee, calculated as a percentage of the selling price, is imposed by the auction house, which is paid by the buyer
CATA	Computer-Aided Text Analysis – a frequency count of words appearing in free-form item descriptions

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Corporate social responsibility	Corporate Social Responsibility (CSR) is when companies go beyond profit by addressing social and environmental concerns in their operations and stakeholder interactions
Herding bias	Herding bias is the tendency to follow group decisions instead of independent judgment due to the desire to fit in and feel secure
Linguistic vagueness	Refers to language that softens our expressions in a way that readers often prefer because vague expressions do not appear unduly direct or assertive
Quality rhetoric	Words used to convey a sense of higher quality for an item
Tone	Words conveying positive versus negative themes
Wordiness	The total number of words used to describe the item

1 Introduction

Online auctions have become an increasingly prominent channel for selling goods [1, 2]. Despite their convenience and reach, these platforms face a persistent challenge: legitimacy. Unlike traditional retail or branded e-commerce environments, online auctions often involve one-off exchanges between anonymous parties and lack enduring reputational mechanisms such as brand recognition, relationship familiarity, or verified quality assurance [3, 4]. In such transient, information-asymmetric contexts, buyers must infer trustworthiness from limited cues available in sellers' listings. Understanding how these cues shape, and sometimes erode, consumer trust remains a critical issue for theory and practice.

A central challenge for sellers in online auctions is earning trust in one-time transactions, because online auctions generally have more informational asymmetries compared to traditional retail channels [5]. Trust provides psychological assurance, motivating consumers to act despite uncertainty [6]. Yet without brand reputation, relationship history, or quality verification, establishing legitimacy becomes difficult [7]. Consequently, scholars have urged greater examination of rhetoric as a substitute for these absent mechanisms [8]. Evidence from adjacent digital settings supports rhetoric's influence on trust formation. For instance, message tone and readability shape funding decisions in credit markets [9], positively framed mission statements increase donations in nonprofits [10], and emotionally charged narratives accelerate entrepreneurial funding [11]. However, auction rhetoric has received limited attention, particularly regarding how certain linguistic strategies may backfire.

The Elaboration Likelihood Model (ELM) is widely used in marketing and consumer psychology to explain how individuals process persuasive messages through central or peripheral routes [12, 13]. While ELM clarifies how consumers interpret rhetorical messages, it offers limited insight into why sellers rely on rhetorical claims or how such efforts might undermine the legitimacy they seek to build. Legitimacy is particularly critical in online auctions because, unlike traditional persuasion settings where messages are reinforced by brands or long-term relationships, auctions lack stable trust mechanisms [3, 14, 15]. From the perspective of the ELM, a brand can

be understood as an amalgamation of the various message variables that serve as persuasive cues supporting a product. In this sense, the brand itself functions as a single, overarching persuasive element that encapsulates and influences the impact of other signals [16]. However, in the case of online auctions, brand information is commonly not available to potential bidders. This raises the question of whether other constructs act as meta-cues in online auctions where brand information is deficient. In line with this, recent scholarship has called for integrating ELM's focus on message processing with trust-formation mechanisms relevant to social and market exchanges [17].

Institutional theory complements this psychological perspective by emphasizing legitimacy and trust as foundations of economic behavior [18, 19]. Firms and individuals gain legitimacy by conforming to social norms and mimetic pressures, reducing uncertainty and information asymmetry [20–22]. When conformity to established norms is not possible, actors often pursue alternative paths to legitimacy through symbolic or rhetorical signaling [23]. Online auction sellers face precisely this condition. Without established reputations or repeated interactions, they must project trustworthiness through alternative cues.

One such approach involves borrowed legitimacy, in which sellers align their credibility with that of others or with socially accepted signals [24]. In illicit online markets, for instance, sellers rely on escrow intermediaries to borrow trust from the third party [25]. In auctions, a similar effect emerges through herding bias, where buyers infer quality and reliability based on the actions of others. This dynamic has been observed in peer-to-peer lending [26], product review contexts [27], and consumer decision-making guided by social cues [28]. Herding bias signals, such as the number of bids an auction item has received, convey social proof that can indicate legitimacy or popularity [29, 30]. Herding bias has implications for both buyers and sellers. For bidders, it can be detrimental because increased competition raises final prices and reduces consumer surplus, whereas for sellers, it enhances the likelihood that an item will sell [5]. The accumulation of bids, therefore, signals legitimacy by reflecting collective endorsement.

Similarly, the length of an item's description functions as a signal of the seller's effort, competence, and transparency [31]. The characteristics of a listing's description provide critical information that helps buyers mitigate information asymmetry, while for sellers, a carefully constructed description can attract interest and stimulate bidding activity. Although both cues reduce uncertainty and encourage participation, they differ in controllability: sellers can directly shape the rhetoric of item descriptions, whereas herding bias depends on the collective behavior of bidders. Additionally, rhetorical institutionalism, a subset of institutional theory, cautions that excessive self-promotion can erode trust [32, 33]. When sellers attempt to project authenticity or quality too assertively, buyers may interpret these claims as insincere, a phenomenon described by Ashforth and Gibbs [34] as "protesting too much." Research in marketing and communication supports this view, showing that overstated or incongruent rhetoric can trigger skepticism rather than confidence [35–37]. Thus, rhetoric functions as a double-edged mechanism: moderate use can enhance credibility, while exaggeration can damage it.

To address this, we combine insights from rhetorical institutionalism and ELM to explain how sellers' rhetorical efforts to gain trust can sometimes backfire. Prior

research highlights rhetoric's positive role in fostering trust, emphasizing value-consistent or emotionally resonant messages. For example, auction sellers can build trust by emphasizing corporate social responsibility [38, 39], utilizing inward-focused usernames that convey positive affect [8], or adopting an optimistic tone [40, 41]. However, little is known about how rhetorical overexpression—such as excessive quality claims or vague persuasion—might erode perceived credibility.

By integrating ELM and institutional theory, we provide an evolved understanding of persuasion and trust formation in online auctions. We theorize that two core listing characteristics—herding biases, indicated by the number of bids, and wordiness, measured by the amount of descriptive text—interact with rhetorical expression to shape auction performance. We focus on two rhetorical dimensions prevalent in online listings: quality rhetoric, which highlights the superiority or authenticity of an item, and linguistic vagueness, which relies on imprecise or inflated statements. While balanced quality rhetoric may enhance trust, excessive or vague rhetoric is expected to reduce trust by increasing perceptions of insincerity [42]. In this way, rhetoric moderates how informational and social signals influence buyer decision-making.

Our model builds upon both psychological and institutional foundations. It posits that restrained, credible rhetoric strengthens the positive effects of social and informational cues on buyer trust, while vague rhetoric weakens these effects. Accordingly, quality-related and vague rhetoric operate as conditional mechanisms that can either reinforce or undermine legitimacy in digital marketplaces. A central motivation for this study is to address a gap identified in prior auction research, which notes that the mechanisms underlying auction success remain insufficiently understood [43]. Specifically, we seek to determine whether certain rhetorical strategies employed by sellers enhance the likelihood that an item will successfully sell.

To test these ideas, we analyze large-scale, web-scraped data comparing online auction items that sold successfully with those that did not. This approach answers the call by Guyt et al. [44] to expand the use of digital trace data in retailing research. By integrating ELM and rhetorical institutionalism, this study advances understanding of persuasion, trust, and legitimacy in online auctions. The findings provide theoretical insight into how communication style interacts with social proof and information cues, as well as practical guidance for sellers seeking to communicate effectively and credibly in high-uncertainty market environments.

2 Literature review of psychological and institutional perspectives and hypotheses development

Digital auctions possess distinctive characteristics that may introduce consumer decision-making biases. Auction sellers collaborate with an online auction platform, which functions as a broker, to list an item for sale [45]. These listings typically feature a written item description, photographs, estimated price, duration, and any applicable reserve price that must be met (if the seller opts to disclose it). A key distinction between online and traditional auctions is that online bidders are unable to physically inspect items, relying exclusively on the information provided in the listing [5]. Consequently, connecting bidders with auction items is more challenging

in the online context, which makes understanding the psychology and socio-institutional aspects of purchase decisions in the auction context an important scholarly and practical endeavor.

2.1 Antecedents of the likelihood of sale in an online auction

Prior literature identifies several key antecedents related to online auction listings that shape the likelihood of an item selling. These include price-related factors (starting bids, reserve prices, buyer premiums), product visibility, time pressure, reputation cues (seller ratings, reviews, and brand indicators), and social proof mechanisms such as herding bias [31, 43]. Additionally, the length and quality of item descriptions—often interpreted as a proxy for seller transparency—can influence perceived credibility and transaction success [46, 47]. While these determinants are individually well-studied, limited research examines how rhetorical properties of language—the style and framing of these cues—affect buyer responses. Our study extends this line of inquiry by examining how rhetorical elements moderate established antecedents of auction success.

Some antecedents of item sales may possibly engage either central or peripheral processing, depending on which aspects of the message are emphasized. Within the framework of the ELM, more objective indicators such as starting bid prices, seller ratings, and brand information are likely processed through the central route, as they require thoughtful evaluation and cognitive effort. In contrast, cues such as time pressure, herding bias, and item descriptions may be more likely to be processed peripherally by buyers, relying on heuristics rather than systematic analysis.

2.2 The psychological perspective in online auctions

Psychological perspectives are highly relevant to understanding buyer decision-making in online auctions because they explain how consumers process information to form trust. Among persuasion frameworks, the ELM has been particularly influential in e-commerce research [13, 48–50]. ELM views consumers as skeptical decision-makers who are persuaded through two distinct processing routes: a central route, which involves deliberate and systematic evaluation of substantive information, and a peripheral route, which relies on heuristic cues that require less cognitive effort [13, 48, 51]. These dual pathways are particularly applicable to digital marketplaces, where buyers often face information overload and vary in their motivation or ability to process dense product information.

Research consistently shows that both routes operate simultaneously. For example, in crowdfunding settings, investors consider central cues such as venture quality and innovativeness while also responding to peripheral cues, including readability, source credibility, and third-party endorsements [12, 52]. Similarly, in live-streaming commerce, both product demonstrations (central cues) and popularity indicators or barrage comments (peripheral cues) influence purchase decisions [13, 53]. Studies of online agricultural markets reveal comparable patterns: buyers respond to central cues such as third-party product quality ratings and to peripheral cues such as seller reputation and positive word-of-mouth [54, 55]. Collectively, these findings suggest

that consumers in digital markets rely on a combination of substantive and heuristic information to build sufficient trust for purchase intentions.

Overall, psychological frameworks such as ELM clarify how buyers navigate the abundance of information and information asymmetry that define e-commerce transactions. Because buyers cannot physically inspect products or directly engage with sellers, they must rely on seller-generated descriptions and other observable cues—such as the number of bids, textual tone, and linguistic style—to infer credibility [48, 53]. Importantly, psychological research shows that information cues often interact, amplifying each other's persuasive effect when they collectively increase perceived trust or positive emotion [12].

2.3 The institutional perspective in online auctions

Institutional theory provides a valuable framework for understanding how economic actors navigate environments characterized by uncertainty and information asymmetry, both of which undermine trust. In market contexts, institutional theory suggests that transactions require trust built through legitimacy, which is achieved by conforming to the norms, rules, and expectations of relevant social actors [15, 56]. Legitimacy represents a perception of appropriateness and trustworthiness that enables exchange. Although institutional theory is often applied at the organizational level, legitimacy pressures also operate at the individual level. Sellers in online auctions, for example, must persuade skeptical buyers—consistent with insights from the ELM—of their trustworthiness.

Efforts to extend legitimacy can be problematic, particularly in contexts where audiences are skeptical. Online auctions illustrate this difficulty because they differ from traditional retail settings in several ways. First, they contain an abundance of information, including item features, options, and intra-category details [57]. Second, participants can observe and be influenced by others' behaviors [58]. Third, auctions often lack conventional trust mechanisms such as physical inspection, brand reputation, third-party verification, or repeated transactions [59]. In these conditions, sellers must seek legitimacy through communicative means rather than structural assurances. Prior research shows that when material evidence is limited, actors such as entrepreneurs, nonprofits, and emerging firms use rhetorical and symbolic strategies to signal alignment with institutional norms [32, 60].

In the context of e-commerce auctions, sellers similarly rely on rhetorical devices and heuristic signals to convey conformity and reliability. These strategies reflect a broader institutional dynamic in which actors attempt to manufacture legitimacy when normative conformity is difficult to demonstrate. However, institutional theory also cautions that legitimacy is granted by audiences, not claimed by actors [15]. When sellers overemphasize their credibility, audiences may perceive their efforts as insincere or “too good to be true,” reinforcing skepticism [34, 35].

Institutional theory also identifies mimetic and normative pressures as mechanisms for legitimacy. Mimetic pressures occur when sellers and buyers model the behavior of others to appear credible in uncertain environments [56]. In auctions, this dynamic manifests as herding bias, where bidders infer legitimacy from prior bidding behavior [27]. Normative pressures reflect expectations for appropriate seller behavior, such

as providing clear descriptions, emphasizing quality, and maintaining a suitable tone. Deviating from these expectations may result in withheld legitimacy. Thus, legitimacy in auctions emerges through a combination of mimetic signals (bids), normative adherence (expected rhetoric), and strategic communication (wordiness, quality claims, and vagueness).

Taken together, an institutional perspective highlights that legitimacy and trust are socially constructed outcomes shaped by communication, conformity, and audience evaluation. Whereas psychological frameworks such as ELM emphasize the cognitive mechanisms of persuasion, institutional theory underscores the social and contextual processes through which legitimacy is formed. Online auctions reveal how sellers depend on rhetorical and heuristic strategies to secure legitimacy, yet these same efforts risk undermining credibility when they exceed audience expectations.

While prior studies largely emphasize the positive effects of central and peripheral cues, less is known about how these same cues can diminish trust and reduce the likelihood of purchase. This limitation motivates the integration of institutional perspectives with psychological ones in the present study. We specifically examine how peripheral cues related to the rhetoric of auction listings may negatively moderate the positive influence of peripheral informational cues on sales outcomes. Our theorizing builds on this perspective to explain how wordiness, quality rhetoric, and linguistic vagueness function not only as persuasive cues but also potentially as contested claims in the broader process of legitimacy construction.

2.4 Herding bias in online auctions

The unique characteristics of online auctions described above collectively contribute to a decision-making process prone to biases. Herding bias is the tendency for consumers to gravitate to listings with more bids and give less consideration to listings with fewer bids [43]. With millions of new listings daily, buyers face the impracticality of thoroughly reviewing all products. Individuals often compromise between making informed decisions and minimizing effort [57, 61]. The desire for efficient decision-making diminishes the importance of scrutinizing the wealth of information, leading buyers to depend more on mechanisms like biases and heuristics [57].

Secondly, since digital auctions are social in nature, they involve interactions among participants that extend beyond economic transactions [57, 58]. Bidders observe and are influenced by reference groups in this mediated social space [62]. Reference groups, defined by individuals serving as sources of values, beliefs, attitudes, and behaviors [63], exert influence based on their members' conformity to norms. Consumers using the observation of others' bids to inform their own purchasing decisions through herding-bias is a form of conformity.

2.5 Listing wordiness in online auction descriptions

Online auctions are examples of Akerlof's [64] market for lemons, where relevant information is moderated by sellers. The seller maintains relevant information about the quality of an item and their intentions that are unknown and often unobservable to the bidder, which engenders problems related to information asymmetry [65]. The

resulting information asymmetry creates the space for problems relating to adverse selection [64, 66]. Adverse selection issues occur when rational bidders approach the auction unsure about the claims being made by the seller [67, 68]. Given the heterogeneity of products in online auctions and the structure where sellers and buyers are separated by a digital divide, the problem of information asymmetry is greater than in physical transactions [69]. Bidders, unsure about the claims being made by the seller, are left to contextual or situational markers such as the narrative used to describe an item to determine, for example, the quality of an item or the reputation of the seller.

In an online auction setting, “increased disclosure” takes the form of sellers who provide more thorough item descriptions. Though more thorough item descriptions may suffer from “protesting too much”, a seller can legitimize their information disclosure if they can demonstrate that deviations are costly [70]. Thorough, detailed auction descriptions signal that the seller is invested in the auction process. Kauffman and Wood [31] document a positive association between an auction description’s length and bidders’ willingness to raise their bid price thresholds, attributing this effect to the belief that providing more information reduces the seller’s ability to engage in deceitful and opportunistic behavior.

2.6 Quality rhetoric in online auctions

Rhetorical institutionalism emphasizes language as a central mechanism through which legitimacy is constructed and judged [71]. Although many factors influence legitimacy evaluations, rhetorical perspectives highlight decision-relevant presumptions—the assumptions audiences make about what is appropriate or credible—as particularly influential [72, 73]. In online auctions, sellers reasonably assume that emphasizing product quality is decision-relevant for buyers and often use quality rhetoric to extend legitimacy. Prior research has consistently identified quality as a key determinant of online commerce outcomes [57, 74, 75]. Yet, an important question arises: are listing descriptions truly an effective space for signaling quality, or do excessive quality claims risk being perceived as “protesting too much” in an attempt to appear legitimate?

Because this study focuses on sellers’ attempts to extend legitimacy through rhetorical claims, we examine peripheral cues related to quality. Specifically, we define quality rhetoric as linguistic emphasis on an item’s excellence using terms such as “sturdy,” “foolproof,” or “resilient” [76], rather than more objective, substantiated measures of quality. How information about quality is processed by potential buyers ultimately depends on what the message emphasizes and whether the claims are presented as objective evidence or as affective, unsubstantiated appeals. When messages emphasize objective, evidence-based, or substantively verifiable aspects of quality—such as factual information, third-party assessments, performance data, or detailed craftsmanship—these constitute central cues, because they require cognitive effort to evaluate the merit of the arguments presented within the specific context [51]. For example, in food labeling, information about protein content, organic certifications, food safety certifications, or lab tests constitutes central cue quality information [77–79]. However, information that signals quality without giving substantiated information such as claims that emphasize “premium”, “chef’s choice”, “made with

love”, “authentic recipe”, “gold label”, “trusted for over 40 years”, and “best on the market” constitute peripheral quality information [77–79]. In such cases, the word “quality” functions as a heuristic or affective signal rather than as evidence, and persuasion arises from positive associations rather than systematic evaluation in line with peripheral processing [51, 80]. Thus, objective demonstrations of quality engage the central route, while unsubstantiated claims about quality trigger the peripheral route of persuasion. In the present study, the measure of quality rhetoric is based on the latter form of expression—keywords that communicate quality heuristically—indicating that, within this context, quality rhetoric represents a peripheral cue.

From a practical perspective, suggesting that sellers increase bids to enhance sales would be tautological. Although herding bias is well established, it offers little actionable insight for sellers seeking to influence buyer behavior directly. A more practical question is how sellers can encourage buyers to respond positively to listing content rather than to herding signals. One strategy highlighted in prior research involves using language to signal product quality [43]. In the absence of traditional selling mechanisms, rhetoric becomes a primary means for sellers to communicate quality [42]. However, quality rhetoric is inherently subjective and nonbinding—it imposes no cost or obligation on the seller [81, 82]. Because such statements are inexpensive and unverifiable, they have been described as “cheap talk” [83].

While cheap talk can be persuasive [84], consumers tend to be skeptical of quality claims that lack objective evidence [85]. Research also suggests that competitive pressure in auctions may incentivize sellers to be honest, yet exaggerated claims can quickly undermine credibility [86, 87]. Thus, although sellers often use rhetoric to promote quality, they must do so judiciously. Overly assertive or inflated claims can heighten buyer skepticism and diminish the effectiveness of other legitimacy signals, such as herding bias and wordiness. In short, restrained and credible quality rhetoric is more likely to foster trust, while excessive claims risk eroding legitimacy and reducing buyers’ receptivity to other cues within the auction environment.

2.7 Linguistic vagueness in online auctions

In addition to what is argued, how arguments are made is also likely to moderate the paths between listing characteristics and the likelihood of selling. Linguistic vagueness refers to language that softens our expressions in a way that readers often prefer because vague expressions do not appear unduly direct or assertive [88, 89], but rather vague language feels conversational and natural [90]. Vagueness is a psychological construct that conveys a communicator’s subtle and indirect admission that they: do not have all the facts; lack a perfect understanding; and are imperfect [91, 92]. Linguistic vagueness incorporates phrasing such as: “sort of”, “fairly”, “not always”, “not quite”, “maybe”, “could be”, and “perhaps” [92]. Sellers may utilize such language in auction listings with the hope of softening the perception that the claims they are making are protesting too much. A seller may convince themselves that it is reasonable to presume that their subtle admission of imperfect knowledge in an auction listing through vague language can lend perceived virtue, through positive emotions to the listing [93, 94]. Most people tend to speak and write with vague lan-

guage in their own lives, and reading an auction listing with vague language is more likely to feel familiar [93, 95, 96].

However, the vagueness of the wording in an item listing can potentially influence the effectiveness of the legitimacy garnered from herding bias and wordiness, which focuses on the performative aspect of language rather than the substantive content of language. Rhetorical institutionalism emphasizes that the performative aspects of communication can shape legitimacy judgments [71, 73]. Communication is critical in advertising, especially in auctions that lack traditional mechanisms to garner trust [97]. A seminal paper in advertising suggests that when consumers are presented with unambiguous evidence, consumers' judgments "are dependent only on the objective physical evidence and unaffected by advertising" [98, p. 221]. Contrarily, when physical evidence is not presented in an unambiguous manner, consumers rely more on whatever information is provided in advertisements to form their purchase decisions [98]. These findings suggest that the degree to which objective information is presented ambiguously versus unambiguously determines whether consumers will attenuate information in advertisements or not. Indeed, this finding spurred auction specific research on how ambiguously specific signals are presented, such as: price [99]; product experience [100]; and quality [101]. However, research on vagueness is sparse and has not yet considered what impact the degree of general ambiguity in rhetoric within listing information has on consumers' purchase decisions [93]. In a related study, Lee and Shapiro [42] study the effect of sellers deliberately trying to conceal their low quality as they compete against higher quality sellers. Although such deceit may be conceptually related to ambiguity and vagueness, the construct that we focus on is general vagueness in language and rhetoric, not the deliberate concealment of important information.

3 Hypotheses development

3.1 Herding bias and the likelihood of selling an item

The mechanism underlying herding bias is termed informational influence [102, 103]. Bidders rely on others' bids not just for convenience but also because they believe it leads to informed purchase decisions. In auctions characterized by pronounced information asymmetry, bidders rely on contextual signals to infer product quality and seller credibility [104]. This information gap helps explain Dholakia et al.'s [43] finding that bidders often view other bidders' behavior as more credible than sellers' descriptions. The credibility of an item increases with the number of bids, indicating a larger influential reference group [43]. So, sellers are starting from a disadvantaged presumption of lack of trustworthiness and legitimacy. Because consumers are suspicious of sellers' information claims, consumers look to others' actions (peripheral cues) to inform their legitimacy beliefs. In short, listings borrow legitimacy to garner trust from a potential buyer from other potential buyers' bids. Consistently, we predict that listings with more bids will have a greater likelihood of selling, reflecting the impact of an influential reference group on actual purchase decisions:

H1 *The greater the number of bids a listing has, the more likely that listing is to sell, consistent with herding bias mentality.*

3.2 Item description wordiness and the likelihood of selling an item

Despite the finding of a positive association between an auction item description length and bidder utility [31], there is literature that suggests that the association is not linear. Research on information overload in online commerce suggests that the vast amount of information presented to buyers can lead to cognitive overload. A review of the information overload literature suggests that excessive information can impair the decision-making process. Overload can lead to longer decision times, a greater probability of errors, and overall decreased satisfaction [46]. Overload also leads to more impulsive and less optimal decisions in the auction process [105]. Hu and Krishen [47] generalize this result and suggest that overwhelmed bidders experience decreased satisfaction and more buyer's remorse. Conversely, concise communications have been shown to positively influence the engagement of audiences [106], supporting the value of brevity when communicating with potential buyers. Relatedly, research has shown that the emoted level of happiness of online influencers has an inverted U-shaped relationship with sales, reinforcing the suggestion that overselling to buyers reaches an optimal point, beyond which there are diminishing returns [107]. Taken together, the literature on asymmetric information combined with information overload suggests that the link between an auction item description length and an auction item's likelihood of selling follows a non-linear path. Thus, we state the following hypothesis:

H2 *The relationship between the wordiness of an item's auction listing and the likelihood of the item selling follows a non-linear, concave-down inverted U-shaped pattern, indicating the existence of an optimal level of wordiness.*

3.3 Rhetorical moderation

The main effects theorized in this study suggest an optimal condition of a listing having many bids and a description with sufficient length to communicate important item details, but not so much length that the description might be perceived as "protesting too much". Under such conditions, our hypotheses predict that buyers are likely to form positive legitimacy judgments about the listing and exhibit greater willingness to bid. Integrating the ELM with rhetorical institutionalism provides a richer account of how these judgments evolve. From an ELM perspective, herding bias and wordiness function as signals that guide buyers' persuasion processes under conditions of limited attention and information asymmetry [48]. Whereas the rhetorical view of institutional decision-making explains how decision makers weigh the risk of viewing information as legitimate versus the risk of doubting [73]. Buyers will want to avoid the negative consequences of being wrong in either incorrect positive legitimacy beliefs or incorrect negative legitimacy beliefs. Rhetorical institutionalism suggests that as people face these two risks, they ultimately form their beliefs and decisions based on argumentation [73, 108]. So, while information about

how many other people have bid on a listing and how thorough the listing is provide useful peripheral signals to help form initial perceptions of legitimacy, potential buyers' positive legitimacy beliefs could be switched to doubt depending on how effective sellers' arguments (or appeals) are [109]. Argumentation can refer to both what is argued and how arguments are made (pathos, logos, and ethos can all be relevant) [73]. To capture these rhetorical dynamics, we theorize that two forms of argumentation—quality rhetoric, representing what is argued, and linguistic vagueness, representing how it is argued—might act as meta-cues in the process of extending legitimacy. These rhetorical elements shape how buyers integrate institutional signals with cognitive persuasion routes, ultimately influencing whether trust is reinforced or eroded in the auction context [16, 110].

3.3.1 The moderating effect of quality rhetoric

Quality rhetoric refers to the proportion of words in an item description emphasizing peripheral claims about quality (e.g., “sturdy”, “reliable”, “premium”.) Beyond its direct rhetorical meaning, quality rhetoric functions as a contextual cue that alters how buyers interpret other legitimacy signals [16, 75]. Within the ELM framework, quality rhetoric does not itself represent substantive information about the product (central-route content) but rather shapes the processing route through which buyers evaluate all subsequent cues [12]. Extensive references to quality can prompt a shift from relatively trustful peripheral processing of social signals (such as herding bias) toward more effortful and skeptical central processing, where claims are scrutinized for inconsistency [85]. From a rhetorical institutionalism perspective, this occurs because overt quality claims signal a defensive attempt to manufacture legitimacy, thereby reframing other communicative signals within a context of potential overclaiming [82]. As a result, the perceived credibility of otherwise positive cues declines. In this way, quality rhetoric operates as a moderator: it does not directly determine sales outcomes, but rather conditions how buyers interpret and integrate herding and informational cues when forming legitimacy judgments [73]. Therefore, due to the risk of excessive quality rhetoric being viewed as cheap talk, we predict that quality rhetoric will negatively moderate the influence that herding bias and wordiness have on the likelihood of an item selling. First, we posit that the more quality rhetoric is emphasized in a listing, the less positive impact herding bias will have on an item selling:

H3a *The degree of quality rhetoric in a listing will negatively moderate the positive relationship between herding bias and the likelihood of an item selling.*

Similarly, long descriptions expounding upon quality are likely to be perceived as “cheap talk” and have the opposite of a seller's intended effect. Institutional theory suggests that resources, confidence, trust, and support flow towards constructs and situations that provide certainty and order by conforming to contextual norms [20, 21, 111]. Additionally, institutional theory distinguishes between maintaining existing levels of legitimacy and trust and extending legitimacy and trust [34]. A well-respected brand, for example, may be able to make frequent proclamations about

quality without the same risk of being perceived as protesting too much that will result when such claims are made by a seller where traditional trust mechanisms are not available. In an online auction setting, quality rhetoric is more in line with extending legitimacy than maintaining legitimacy, and therefore, such efforts are more likely to be scrutinized and viewed suspiciously [112]. Hence, we expect that a higher degree of quality rhetoric will negatively moderate both of the main effects:

Building on our earlier hypothesis that the relationship between the wordiness of an item's auction listing and the likelihood of an item selling follows a non-linear, inverted U-shaped curve, the introduction of quality rhetoric as a negative moderator will manifest by lowering the optimal level of wordiness. Specifically, quality rhetoric is expected to diminish the point at which increased wordiness maximizes the probability of an item selling, thereby shifting the peak of the curve to the left. Thus, we predict:

H3b *The degree of quality rhetoric in a listing will negatively moderate the concave-down relationship between the wordiness of an item's auction listing and the likelihood of the item selling, such that for items with listings having higher levels of quality rhetoric, the maximum likelihood of an item selling and the optimal levels of wordiness are lower.*

3.3.2 The moderating effect of linguistic vagueness

From an institutional theory perspective, the fundamental hurdle for consumers positively attenuating sellers' claims contained within a listing is a lack of trust due to deviance from norms [71]. While vague language may convey familiarity, virtue, and positive affect in some contexts, in a situation where extending legitimacy is critical, vague language may also be problematic because it represents deviance from norms. Humans have a natural tendency to be suspicious of deviance from norms in an effort to extend legitimacy [34]. Linguistic vagueness in auction settings may exert the opposite moderating effect from the softening observed in other contexts, as such vagueness further distances a listing from what consumers perceive as normative, and in online auctions, buyers' evaluations typically begin from a baseline of suspicion. As noted, auction settings lack traditional mechanisms of brand name, repeated relationships, quality through reputation, and the like [3, 4]. Consumers are likely to be particularly predisposed towards listings that provide such information as matter-of-factly as possible. Although linguistic vagueness does not mean intentionally withholding such information, linguistically vague rhetoric is likely to convey a feeling of uncertainty [91, 92]. When meaning is not conveyed in an objective and direct manner, readers are left to assign some meaning themselves. A growing body of research on consumer psychology specifically suggests that consumers are risk-averse and tend to be skeptical [113]. So, when meaning is left to be ascribed or interpreted by the consumer because an auction listing is linguistically vague, consumers are more likely to ascribe a skeptical interpretation.

Beyond its independent rhetorical meaning, linguistic vagueness alters how buyers process and interpret other legitimacy signals within an auction listing. In the ELM framework, vague phrasing acts as a meta-cue that changes the route of persua-

sion: by lowering message clarity, it reduces the cognitive elaboration buyers devote to other cues [114]. As a result, when listings are highly vague, buyers may rely more on heuristic judgments (such as the presence of bids) but also treat these heuristics with caution, perceiving them through a lens of uncertainty. From a rhetorical-institutional perspective, linguistic vagueness signals an indeterminate or non-committal communication style that modifies the meaning of accompanying information [15, 115]. When vagueness accompanies clear social cues such as herding bias, it may erode the credibility of those cues by implying inconsistency or evasiveness. Thus, linguistic vagueness operates not as a direct determinant of sales outcomes, but as a moderator that conditions how buyers interpret informational (wordiness) and social (herding) signals when forming legitimacy judgments. Specifically, we hypothesize that linguistic vagueness moderates the effects of both herding bias and wordiness on the likelihood of an item selling, weakening the positive influence of herding bias while potentially altering the curvature of the wordiness–sales relationship:

H4a *The degree of linguistic vagueness in a listing will negatively moderate the relationship between herding bias and the likelihood of an item selling.*

H4b *The degree of linguistic vagueness in a listing will negatively moderate the concave-down relationship between the wordiness of an item's auction listing and the likelihood of the item selling, such that for items with listings having higher levels of vagueness, the maximum likelihood of an item selling and the optimal levels of wordiness are lower.*

3.4 Summary of hypotheses

Prior investigations of auction literature and related studies lead us to predict that herding bias effects will increase the likelihood of an item selling in an online auction. In contrast, we expect there to be an inverted U-shaped relationship between the wordiness of an item listing and the probability of an item selling. We also anticipate that these effects will both be negatively impacted when item listings contain exces-

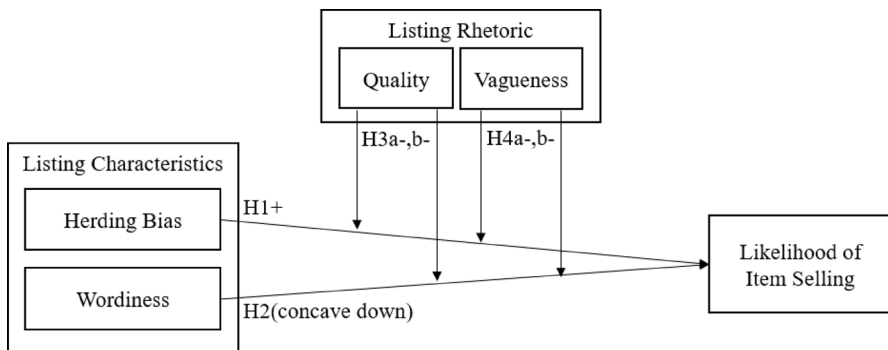


Fig. 1 Research model

sive levels of quality rhetoric or linguistic vagueness. The model summarizing the theorized relationships and our hypotheses is depicted in Fig. 1.

4 Data and method

4.1 Sample

We obtained our dataset by utilizing a custom-designed application programming interface (API) web scraping tool to extract auction data from LiveAuctioneers.com. As noted by Guyt et al. [44, p. 133], a “crucial underutilized benefit of web scraping is that it lets researchers examine phenomena unobtrusively, which is difficult with more established methods.” LiveAuctioneers was founded in 2002 and is the world’s leading platform for live and timed auctions of collectibles, antiques, and fine art [116]. They note that there are over 5,000 auctioneers and other sellers who post items on their site. For our data collection, auctions that ended in years 2003 to 2022 were captured. The initial sample comprised 243,451 completed auctions. We case-wise deleted auctions that were missing variables used in this analysis, resulting in a final sample of 230,270 auctions for products across 8 categories. Of the auctions in the sample, 205,862 (89.4%) ended with an item being successfully sold. The remaining auctions failed to result in a sale because the reserve price requirement was not met. Table 1 shows the number of items in our dataset by category. Art items had the most items auctioned in this time period, while Furniture had the least.

4.2 Explanatory variables

The numerical measures used in our analysis were extracted directly from each auction listing. To construct the quality rhetoric, linguistic vagueness, and tone variables used for the content analysis, the free-form text field (Item Description) in each auction was analyzed using computer-aided text analysis (CATA) to count occurrences of keywords within a listing. This and similar text mining approaches have been shown to be effective methods of extracting conceptual content from free-form text data [117–120]. Although less sophisticated than advanced approaches such as machine learning-based natural language processing in capturing higher-order contextual and semantic dimensions of narrative text [121], CATA has nonetheless been repeatedly recognized as a robust and effective method for the reliable identification

Table 1 Auction data by item category

Category	# of items
Antique& vintage art	45,411
Collectibles	47,033
Fashion	20,853
Furniture	23,972
Home & decor	21,335
Jewelry	39,741
Other	31,925
Total	230,270

of rhetorical patterns [122], which represent the primary variables of interest in this study. To construct our measures, we drew on prior consumer research and online auction studies [e.g., 9, 38, 123, 124] to identify established word-list dictionaries of quality-related terms [76], vagueness-inducing terms [125], and tone-related terms [126], each of which was developed and validated by its original creators.

The variables included in our analyses, as well as the sources of the keywords and terms used in the text mining process are presented in Table 2. The natural logs of two variables, herding bias and wordiness, are used as it was discovered that they were not normally distributed. Additionally, before conducting our analysis, we centered and standardized these measures, which allows for a like-to-like comparison of these

Table 2 Overview of explanatory variables

Variable	Description
Dependent variable	
Item sold	A binary variable set to 1 if the item sold by the end of the auction, 0 otherwise.
Independent variables	
Herding bias	Measured as the number of bids an item received. Herding bias can influence whether an item sells [129]; however, having bids does not necessarily mean that the item received a high enough bid (beyond the reserve price) to be sold.
Wordiness	The wordiness of a listing is assessed using the total number of words in an item description. The squared value of the term is also utilized to determine the concavity of the relationship with the likelihood of selling.
Quality rhetoric	This refers to the percentage of words in an item listing that match words typically used to describe aspects of quality [76]. Sample words include: COMPLIANT, EFFICIENT, FOOLPROOF, RESILIENT, STURDY.
Linguistic vagueness	The percentage of words in the item listing that match words from the Communication Vagueness dictionary created by Hiller et al. [92, 125]. Sample words include: A LITTLE, APPEARS, MAYBE, PERHAPS, SORT OF.
Control variables	
Buyer premium	The Buyer's Premium is an additional service fee, calculated as a percentage of the selling price, imposed by the auction house, which is paid by the buyer [130].
Estimated low and high selling prices	For each item, the seller estimates both the minimum and maximum selling values. Previous research has demonstrated that these estimated values significantly influence bidder behavior and affect the probability of a successful sale [131].
Starting bid price	The starting bid price, which is the minimum bid that will be accepted during the auction, is included as a control as it has been shown to influence the number of bidders participating in the auction (i.e., Herding Bias) [132].
Tone	Studies have identified that the tone utilized to present information in online contexts can influence participants [9]. Using Henry's [126] positivity and negativity word lists, tone is equal to the count of positive words minus the count of negative words, divided the result by the total number of positive and negative words combined.
COVID	A binary variable set to 0 if the auction end date was prior to March 11, 2020, otherwise it is set to 1.
Recession	A binary variable set to 1 if the auction end date was between December 2007 through July 2009, otherwise it is set to 0.
Category	The items in our dataset are classified into eight distinct categories. To identify the category to which each item belongs, seven binary variables were utilized.
Year	Nineteen binary variables are used to control for the year in which auctions occurred.

Table 3 Variable descriptive statistics and correlations

Variable	Mean	Std. Dev.	1	2	3	4	5	6	7	8	9	10	11
1 Dependent variable	0.894	0.308											
2 Item sold													
Independent variables													
2 Herding Bias	8.25	9.47	0.485										
3 Wordiness	56.22	75.93	-0.012	0.052									
4 Quality rhetoric	1.2%	3.5%	-0.011	-0.009	-0.062								
5 Linguistic vagueness	1.6%	9.5%	0.001	-0.006	-0.18	0.036							
Control variables													
6 Buyer premium	22.5%	6.3%	0.248	0.215	0.042	-0.074	-0.020						
7 Estimated low selling price	\$1,492	\$3,525	0.010	0.068	0.144	-0.002	-0.018	-0.010					
8 Estimated high selling price	\$2,796	\$5,802	0.041	0.103	0.095	0.000	-0.019	-0.025	0.827				
9 Starting bid price	\$644	\$1,765	0.008	-0.030	0.121	-0.009	-0.016	0.087	0.738	0.628			
10 Tone	0.306	0.914	-0.010	0.032	0.263	0.016	-0.031	-0.023	0.052	0.055	0.017		
11 COVID	0.489	0.500	0.057	0.178	0.092	0.020	0.013	0.176	0.065	0.082	0.042	-0.026	
12 Recession	0.007	0.085	0.013	-0.151	-0.024	-0.004	0.021	-0.043	-0.017	-0.020	-0.014	-0.010	-0.084

measures' regression coefficients [127]. The descriptive statistics and correlations for all measures are reported in Table 3.

4.3 Methodology

Given the binary characteristic of the dependent variable, we opted for logistic regression as the appropriate analytical method [128]. Our investigation employs three distinct models to scrutinize four hypotheses. Model 1 focuses on the control variables. Model 2 assesses the primary effects of Herding Bias (H1) and Wordiness (H2) on the probability of an item's successful sale. The final model, Model 3, integrates the interactions between quality rhetoric and linguistic vagueness with the main effects (H3a, b and H4a, b).

5 Results

We used Stata 17 to run the logistic regression analyses, using the binary variable sold as the outcome variable. The results are shown in Table 4. Note that the category and year control variables are suppressed for parsimony. Model 1, which examines our control variables, shows that five of the seven variables significantly relate to the likelihood that an auction item will successfully sell ($p < 0.01$ for four variables and $p < 0.1$ for one measure). Of note, the control variable starting bid price was also positive and significant, indicating that the minimum allowed opening bid influences the likelihood of an item selling. A starting bid that is set too low may signal to bidders that the item lacks value, thereby reducing its perceived attractiveness [132].

Model 2 adds in the independent variables to test the main effects of herding bias (H1) and wordiness (H2) (as well as wordiness squared to test for a U-shaped relationship). This model demonstrates that herding bias does positively relate to the likelihood that an item will sell ($p < 0.01$), supporting H1. The significant negative coefficients for both the wordiness and wordiness² terms demonstrate that there is an inverted U-shaped relationship (with an optimal point) between the length of an item description and the probability of an item selling, which supports H2. To better understand this relationship, we present Fig. 2, which graphically shows the relationship between wordiness and the likelihood of an item being sold. As shown in the figure, the optimal length of an item description is approximately 0.5 standard deviations (SD) below the mean level of wordiness in our sample.

Model 3, which includes the interactions of quality rhetoric and linguistic vagueness with herding bias, wordiness, and wordiness² is used to test H3a, H3b, H4a, and H4b. For herding bias, we find that both interaction terms are significantly and negatively related to the likelihood of an item selling ($p < 0.01$). The findings suggest that elevated levels of both quality rhetoric and linguistic vagueness exert a negative moderating effect on the relationship between herding bias and the probability of an item being sold, supporting H3a and H4a. The interactions between quality rhetoric and the wordiness and wordiness² terms are both negative and significant ($p < 0.01$), indicating support for H3b. Contrary to our expectation in H4b, the interaction between linguistic vagueness and wordiness was positive and significant ($p < 0.01$),

Table 4 Hypotheses tests—logistic regression results

Variables	(1) Item sold (yes/no)	(2) Item sold (yes/no)	(4) Item sold (yes/no)
Herding bias (H1)		2.781*** (0.0168)	2.779*** (0.0169)
Wordiness (H2)		-0.0604*** (0.00967)	-0.0503*** (0.0113)
Wordiness ² (H2)		-0.0550*** (0.00635)	-0.0558*** (0.00677)
Quality rhetoric		-0.00219 (0.00904)	-0.143*** (0.0219)
Linguistic vagueness		0.0171* (0.00886)	-0.0135 (0.0391)
Herding bias × quality rhetoric (H3a)			-0.0499*** (0.0139)
Wordiness × quality rhetoric (H3b)			-0.141*** (0.0180)
Wordiness ² × quality rhetoric (H3b)			-0.0339*** (0.00536)
Herding bias × linguistic vagueness (H4a)			-0.0565*** (0.00799)
Wordiness × linguistic vagueness (H4b)			0.101** (0.0405)
Wordiness ² × linguistic vagueness (H4b)			0.0363*** (0.0105)
Buyer premium	0.101*** (0.000952)	0.0537*** (0.00132)	0.0527*** (0.00132)
Est. low price	-0.000154*** (7.07e-06)	-0.000133*** (7.46e-06)	-0.000132*** (7.45e-06)
Est. high price	0.000150*** (5.20e-06)	3.40e-05*** (4.80e-06)	3.31e-05*** (4.79e-06)
Starting bid price	-0.000101*** (6.35e-06)	0.000150*** (8.59e-06)	0.000153*** (8.61e-06)
Tone	0.0132* (0.00790)	-0.0175 (0.0114)	-0.0180 (0.0114)
COVID	-0.0858 (0.0605)	-0.0684 (0.0813)	-0.0620 (0.0813)
Recession	-0.0846 (0.182)	0.0476 (0.182)	0.0595 (0.182)
Constant	-2.060*** (0.155)	3.816*** (0.160)	3.827*** (0.160)
Observations	230,270	230,270	230,270

Standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Hypothesized relationships are shown in **bold italics**.

suggesting that the inclusion of vague or softening language appeared to reduce the skepticism that normally accompanies long descriptions.

Interpreting quadratic interactions solely on the basis of coefficient values is particularly challenging. To enhance our understanding of these relationships, Figs. 3

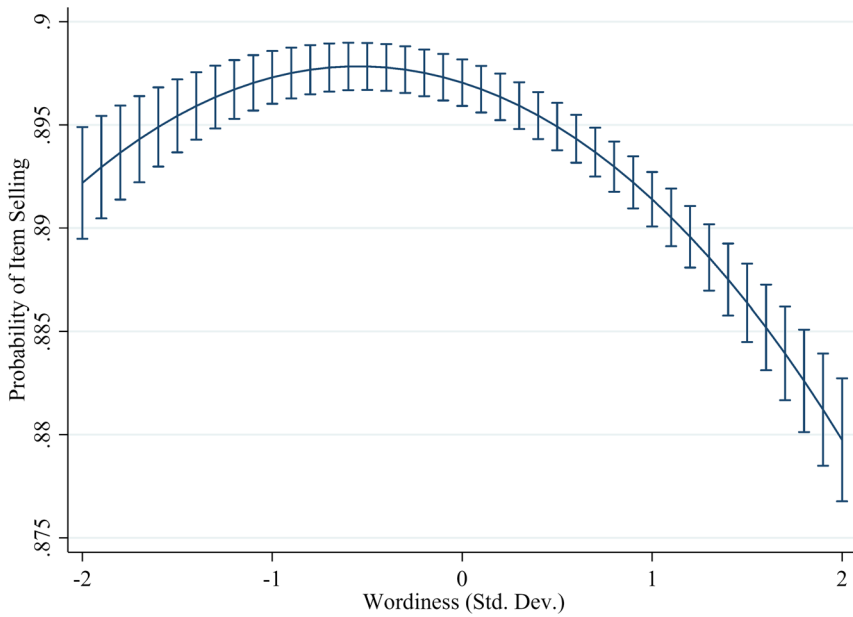


Fig. 2 How wordiness influences the probability of an item selling in an auction

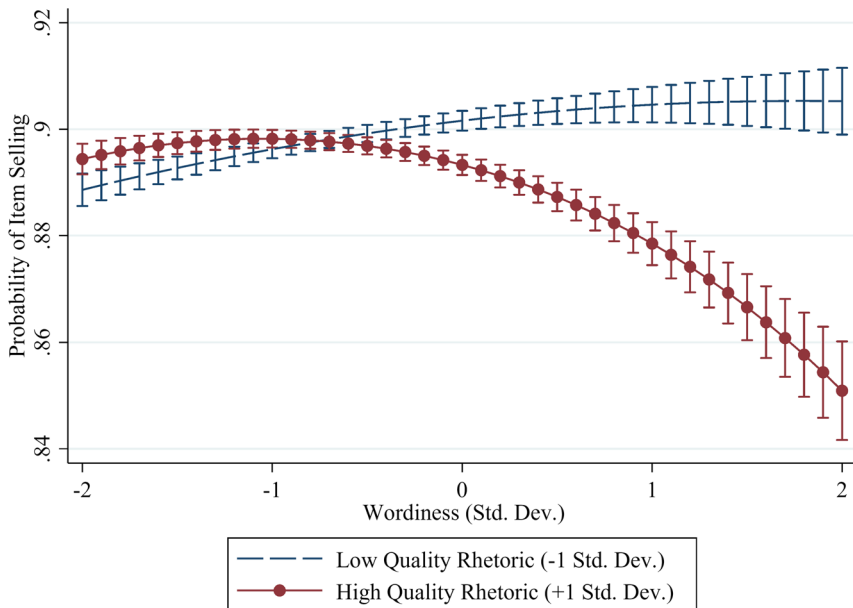


Fig. 3 The negative moderating effect of quality rhetoric on the relationship between wordiness and the probability of an item selling in an auction

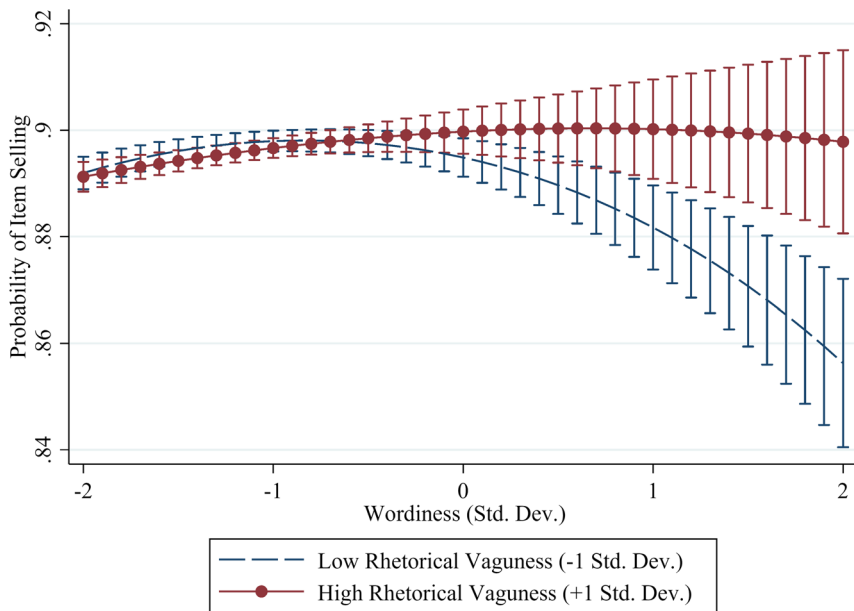


Fig. 4 The positive moderating effect of linguistic vagueness on the relationship between wordiness and the probability of an item selling in an auction

and 4 present visual representations of how the interactions between wordiness and both quality rhetoric and linguistic vagueness relate to the likelihood of an item being sold. The figures also display 95% confidence intervals, thereby illustrating the statistical significance of the differences between low and high levels of the moderating rhetoric variables. Figure 3 demonstrates that the optimal length of an item description is shorter when quality rhetoric is higher. Specifically, when quality rhetoric is high, the optimal length is approximately 1.0 standard deviations (SDs) below the mean, whereas when quality rhetoric is low, the optimal length is about 1.8 SDs above the mean. The confidence intervals further indicate that the difference between the moderating effects of low and high levels of quality rhetoric becomes distinct at roughly 0.4 SDs below the mean level of wordiness. Figure 4 shows that the optimal description length occurs at approximately 1.0 SDs above the mean when linguistic vagueness is high, and at about 1.0 SDs below the mean when linguistic vagueness is low. The confidence intervals indicate that the difference in outcomes between low and high levels of linguistic vagueness does not become significant until description length exceeds approximately 0.9 SDs above the mean. The implications and contributions of these findings are discussed in detail in the following section.

6 Conclusions

This study leverages a dataset comprising over 230,000 completed online auctions to explore the interplay between herding bias, item description wordiness, and two rhetorical attributes of item descriptions: quality rhetoric and linguistic vagueness. Consistent with our hypotheses, we observe that herding biases significantly influence the probability of an item's sale and identify an optimal description length that maximizes that probability. Our findings reveal that elevated levels of quality rhetoric and the use of ambiguous, vague language attenuate the effects of herding bias on the likelihood of a sale. Similarly, our results imply that higher levels of quality rhetoric decrease the likelihood of an item selling as the length of the description increases. Conversely, we find that using higher levels of linguistic vagueness in long item descriptions increases the odds of selling the item. The subsequent sections further discuss these findings and address the theoretical and practical implications of these results, alongside the study's limitations and potential avenues for future research.

The generalizability of this study's findings to other contexts warrants careful consideration, as our dataset was drawn exclusively from LiveAuctioneers.com, a platform that primarily features unique and collectible items. Nevertheless, because most of our results align with prior research conducted across diverse online marketplaces, we expect that certain patterns will extend to other platforms that sell comparable items—for instance, online auctions of similar product genres on eBay. At the same time, additional research is needed to assess whether these findings apply to less distinctive, commodity-like products or to goods offered by firms with established brand recognition and reputational capital. Similarly, extending our findings to non-auction online platforms, such as Amazon.com, necessitates further empirical investigation.

6.1 Theoretical implications

The primary question that this research focuses on is how certain features of auction rhetoric might undermine the legitimacy and trust-garnering efforts of sellers. In other words, we investigate if rhetoric might play a role as a surrogate meta-cue in online auction context, given that the traditional meta-cue of brand is typically absent from that e-commerce context. We find that both listing characteristics (the number of bids and wordiness of a listing) play a direct role in consumer purchase decisions and that listing rhetoric (quality and linguistic vagueness) might undermine sales by moderating both the persuasive trust induced by legitimacy-lending social cues of herding bias and the pervasive cue that an optimal level of information disclosure can have. These findings enable us to make externally valid theoretical contributions related to rhetorical institutionalism and extending legitimacy within online auctions. Importantly, those institutional and legitimacy theoretical contributions increase our understanding of ELM's explanation of how consumers form intentions in e-commerce auction settings where the traditional meta-cue, brand, is absent.

Our study provides alternative meta-cues in the form of rhetoric that sellers should be aware of given their moderating impact on herding bias and information disclosure. Specifically, our contributions highlight how rhetorical elements of listings, such as high emphasis on quality rhetoric and vagueness in their rhetoric, may

shift buyers away from giving credibility to otherwise positive antecedents to online auction sales. The coupling of ELM logic grounded in psychology with rhetorical intuitionism together provides greater understanding of how rhetoric in auction listings might undermine efforts to gain legitimacy and trust. Rhetorical institutionalism helps us understand why gaining legitimacy is difficult even when persuasive information and powerful antecedents, such as herding bias, are present. Specifically, studying rhetoric through an institutional lens highlights why buyers might shift from acceptance of information towards more skepticism if the rhetoric of that information conveys “cheap talk” from an institutional lens. Institutional theory, then, increases our scholarly understanding of why some information that is normally thought of as positive information to convey (such as unsubstantiated, peripheral claims about quality) may be detrimental rather than helpful in forming consumer intentions.

6.1.1 Implications from listing characteristics

Herding bias is an important determinant of sales in online auctions, and scholarly understanding suggests two fundamental reasons for herding bias’s impact. First, converging to listings that already have bids is convenient for buyers [133]. Such informational influence is often considered a fundamental mechanism underlying herding bias [102]. However, the information provided by converging to the norms of other bidders goes beyond just convenience. The second reason herding bias positively impacts sales is that bidders view information provided by other bidders as credible, possibly even more credible than information provided by sellers [43]. The positive impact of herding bias suggests that sellers are starting from a disadvantaged presumption of lack of trustworthiness and legitimacy. Although herding bias is not a new finding in online auction literature, understanding herding bias from an institutional perspective helps emphasize that sellers are likely to engage in rhetoric aimed at extending legitimacy. That reality establishes the importance of the balance of this study’s theorized model of herding bias effects being tempered by protesting too much in auction rhetoric. Specifically, the importance of integrating herding bias and institutional literatures is evidenced by our significant, independent variable that focuses on the verbosity of claims made by a seller to extend legitimacy (listing wordiness), the moderator related to the type of arguments made to extend legitimacy (quality rhetoric), and the moderator related to how arguments are made to extend legitimacy (linguistic vagueness).

Item descriptions act to mitigate the problems of information asymmetry in online auctions. It seems to do so, however, at the risk of information overload. Our findings suggest that buyers may experience cognitive overload when descriptions contain too much detail (i.e., are too wordy.) This insight aligns with prior research, which suggests that excess information leads to impairments in the decision-making process [46, 47, 105]. While detailed descriptions may be necessary to mitigate information asymmetry, there seems to be a threshold beyond which too much information possibly harms decision-making. However, when considered alongside prior online auction research showing that information transparency is not inherently detrimental [31], our findings suggest that identifying the optimal level of wordiness requires balancing information coverage with the risk of information overload. This result offers

a more nuanced understanding of how sellers can strategically manage information disclosure to maximize the likelihood of a successful sale.

6.1.2 Implications from listing rhetoric

Although the positive impact that herding bias has on the likelihood of items selling is well established in the extant literature, this study's identification of theoretical constructs that potentially reduce consumers' reliance on herding bias is an important theoretical contribution. Scholars from a variety of disciplines note the importance of identifying theoretical constructs that reduce reliance on herding [e.g., 134–136]. We theorize and find empirical support for two moderators (quality rhetoric and listing rhetoric) related to listing rhetoric (both under the seller's control when crafting the listing description) that appear to reduce auction consumers' reliance on herding bias.

A rhetorical institutional perspective emphasizes that we tend to argue for what we presume is relevant [73]. It is reasonable for sellers to presume that quality rhetoric is relevant and important to consumers and, therefore, include extensive quality rhetoric within their auction listings. However, rhetorical intuitionism suggests that the credibility of information related to quality is more important than the amount of information related to quality. Because buyers cannot physically inspect goods in an online auction setting and they lack traditional third-party verification methods for quality claims, rhetoric becomes a primary mechanism for sellers to try to extend beliefs about quality [4, 59, 81]. Yet, quality rhetoric does not adequately provide any third-party verification of quality nor does quality rhetoric obligate a seller to meet a minimum standard of quality [82]. Importantly, from a rhetorical institutional theoretical lens then, extensive quality rhetoric is likely to be viewed as “cheap talk”. The more extensive the quality rhetoric is within an auction listing, the higher the likelihood of suspicion about that quality rhetoric. Consumers are often unwilling to merely accept claims of quality that lack some objective verification [85]. Our finding that quality rhetoric negatively moderates the positive impact that listing characteristics can have on an item selling is consistent with rhetorical institutionalism's perspective that efforts to extend legitimacy are likely to do the exact opposite, undermine legitimacy. Importantly, we examine the extensiveness of quality rhetoric. So, the theoretical implication is not that quality rhetoric is universally negative; rather, extensive efforts to emphasize quality that are not substantiated with objective and observable information (e.g., third-party benchmarks or certifications) possibly undermine legitimacy and sales.

Scholars have recently called for a more theoretical understanding of how appeals to pathos specifically influence herding behaviors [137, 138]. Our contributions related to the second moderator, linguistic vagueness, moderating the impacts of herding bias and listing wordiness directly contribute to these calls. Rhetoric literature from a variety of disciplines notes that linguistic vagueness may provide the benefit of emotions related to familiarity and virtue [93, 95, 96]. In regards to herding bias, the intended positive pathos that is presumed to come from linguistic vagueness does not have its normally observed positive impact within online auctions. Rhetorical institutionalism helps explain this finding as it emphasizes that what ultimately influences evaluations from relevant constituents, here potential buyers, is legitimacy

gained by adherence to norms. So rhetorical institutionalism argues that shared language that is consistent with contextual norms is more useful for garnering the trust and legitimacy needed for a sale than linguistics related to pathos, such as linguistic vagueness, if such rhetoric deviates from norms. Indeed, because linguistic vagueness deviates from consumers' familiarity with traditional direct consumption information, humans might be suspicious of such language, especially when that language is perceived as being used to extend trust and legitimacy [34, 71, 139].

The finding that linguistic vagueness increases the likelihood that an item with a lengthier description will sell—contrary to our prediction—suggests that the role of vagueness may be more complex than anticipated. From an institutional theory perspective, we expected linguistic vagueness to be perceived as a deviation from normative expectations, thereby undermining buyer trust [71]. Instead, our results appear to show the opposite effect: vagueness in item listings can create a conversational tone, which may mitigate concerns that a longer description represents “protesting too much” or is otherwise too good to be true. This reversal is theoretically consistent with some research showing that vague language can function as a softening device that reduces the perceived assertiveness or overclaiming inherent in long descriptions. In discourse studies, vague expressions create a conversational, less confrontational tone that reduces the risk that detailed or lengthy communication is interpreted as suspicious or overly promotional [88, 89]. Prior work also finds that vague language promotes positive affect and familiarity, which can mitigate defensive reactions to information-dense messages [94, 95]. From a rhetorical-institutional perspective, vagueness may therefore buffer buyers from perceiving very long descriptions incredulously, reducing the skepticism typically triggered by excessive detail [34, 35]. Within the ELM framework, this result can be interpreted as linguistic vagueness shifting buyers away from effortful counter-arguing and toward lighter heuristic processing when message complexity increases. When descriptions are long, vague language reduces cognitive load [91] and thereby softens the perceived burden of processing dense information. Buyers may thus become less sensitive to the overload effects typically associated with wordy listings [46, 105]. Collectively, these mechanisms might help explain why linguistic vagueness may reverse the predicted moderating direction.

6.2 Practical implications

The findings of this study offer valuable implications for sellers leveraging online auctions to market their goods. Table 5 provides a summary of the key recommendations for sellers to consider when crafting auction item descriptions. The discovery that there is an optimal item description length—beyond which the likelihood of a sale decreases—is perhaps the most directly applicable insight. Considering this finding was theorized to be a result of information overload [105], the observed results may be a case where consumers looking at long descriptions feel too overwhelmed to read and consider purchasing those items. For sellers, knowing that item descriptions need to contain enough words to give salient information to potential bidders, but not be overly wordy, should lead them to use restraint when preparing listing descriptions. In essence, maintaining conciseness in item descriptions proves advantageous.

Table 5 Item description recommendations for sellers

Dos	
Recognize that consumers evaluate your message using substantive information and performance heuristics	Be deliberate in both what you say and how you say it, recognizing that consumers process information both centrally (evaluating substantive information) and peripherally (using cues that may not be directly related to the substantive information about the listing). Consumers often do not inform their decisions based on substantive information alone. Consumers often form intentions based on heuristics (such as evaluating whether or not others are bidding on the same item), tone, and rhetorical elements of the listing. Consider both when crafting listings.
Craft more concise descriptions	Items with descriptions around 50 words in length, which is slightly shorter than the average length observed (56 words), have the overall highest probability of successfully selling.
Use softer language in longer descriptions	If a longer item description is necessary, sellers may benefit from incorporating vague terms that soften the language. The use of less assertive wording can help prevent bidders from perceiving a seller's claims as overly strong, unrealistic, or too good to be true.
Set an appropriate starting bid price	Items with very low starting bid prices exhibit a lower probability of selling. By contrast, starting bid prices that are closer to the estimated selling prices of items likely help convey product attractiveness, thereby drawing in more buyers.
Don'ts	
Be too sparing	Although we observed that descriptions shorter than the average relate to a higher probability that an item sells, the likelihood of an item selling decreases as descriptions get too short or too long.
Be too verbose	Caution is warranted with very long descriptions (exceeding 65 words), as the likelihood of an item selling declines precipitously once this threshold is crossed.
Overemphasize quality in descriptions	The overuse of quality-related terminology in descriptions exceeding approximately 50 words decreases the likelihood of a sale, likely because bidders interpret excessive emphasis on quality as "cheap talk." Be thoughtful about whether the quality rhetoric included is central (verified and objective) or peripheral (unsubstantiated claims) and apply the constraint explained above for peripheral quality rhetoric.
Be too concerned with excessive quality rhetoric or assertive language in shorter descriptions	For shorter descriptions (fewer than approximately 50 words), the levels of quality rhetoric and linguistic vagueness do not substantially affect the probability of an item being sold.

Although this recommendation may seem intuitive, the sellers in our sample did not appear to adhere to it, as, on average, the length of their listings exceeded the optimal length.

The result of the hypothesis test of herding biases, which confirms prior studies that found that an influential reference group can positively impact purchase decisions [43], is not unexpected—logically, items receiving more bids should be more likely to successfully sell in an auction. Of more interest to sellers are the findings of how the effects of herding biases are impacted by the rhetorical characteristics of the item descriptions. Given that sellers do not have direct control over how many bids their listing will get, an important takeaway is that the choice of words used in item descriptions by sellers can affect how the number of bids an item receives influences potential buyers. Specifically, the finding that the benefits of herding bias on the likelihood of an item selling are mitigated when item descriptions contain higher levels

of quality rhetoric should guide sellers when they are composing listing descriptions to be judicious and measured when touting the quality of items. Similarly, the finding that vagueness also negatively affects the relationship between herding bias and the likelihood of an item selling should inform sellers to be precise and unambiguous when crafting item descriptions.

For sellers, the applicability of the findings related to the interactions of quality rhetoric and linguistic vagueness with the wordiness of an item description is more nuanced than the herding bias findings. As the results show, for items with low wordiness (very short descriptions), more emphasis on quality or the use of vague language does not relate to a statistically significant difference in the probability of an item selling. This implies that for items with very short descriptions, the levels of quality rhetoric and vagueness do not appear to make a difference to consumers, which may be driven by the brevity of the description not resulting in information overload. However, as wordiness increases, descriptions with lower quality rhetoric and more vagueness both relate to a higher probability of an item selling. Specific to quality, for average length descriptions, the use of measured versus high levels of quality rhetoric significantly increases the likelihood of an item selling, which aligns with the theoretical evidence that consumers may view excessive quality rhetoric as cheap talk [83]. This difference becomes more substantial as the wordiness of descriptions increases beyond the average length; however, due to the concavity of the main wordiness effect, the probability of an item selling decreases regardless of the level of quality rhetoric. In line with quality rhetoric, vague language can reduce the effect of herding bias on an auction's successful outcome. However, contrarily, increasing the use of vague terms can enhance the likelihood of an item's sale for items with longer descriptions. Given that sellers do not have direct control over the number of bids, but do have the ability to influence the vagueness of an item description, our results should encourage sellers to prioritize and focus on crafting item descriptions that adopt a softer, more conversational tone, taking care not to excessively use vague language. Taken together, these findings suggest that if an item requires a lengthier description, sellers should utilize vague language, incorporating a limited emphasis on an item's quality, to maximize the likelihood that the item will successfully sell.

6.3 Limitations and future research

We recognize several limitations of our study and suggest possibilities for future research. First, although our dataset includes all of the information publicly available on LiveAuctioneers, additional information, such as the buyers' demographic and other personal characteristics (which is not available due to privacy concerns), could enhance the richness of our study. Additionally, unmodeled or unobservable characteristics of LiveAuctioneers may differ from those of other platforms (e.g., eBay) in a way that limits the broader applicability of our findings. Similarly, the distinctive nature of the products sold on LiveAuctioneers and the reliance on an auction-based format necessitate follow-up studies to evaluate the generalizability of our findings to other marketplace contexts. Furthermore, we rely on CATA to assess action item descriptions for rhetorical elements such as quality and linguistic vagueness. While this method enables us to efficiently process large volumes of data, it may impart

certain limitations in capturing the nuances of rhetorical language. Although we recognize this limitation, the use of established rhetoric measures aligns the present research with existing literature to help accumulate our body of knowledge about the influence of rhetoric [140]. Future studies may be able to utilize human subjects evaluation, conjoint or discrete choice experiments, or other more refined approaches to better capture the subtleties in language. Moreover, the items on LiveAuctioneers are all physical goods. Our findings may not generalize to auctions for digital goods, services, or purchases that require significant buyer/seller interaction (e.g., real estate transactions). Future studies could expand our results into these other auction environments. Finally, a potential limitation of this study pertains to the possibility of endogeneity affecting our findings. This concern is inherent in research where unobserved variables may exert an influence on the observed outcomes. However, we contend that there is no compelling theoretical rationale to suggest that endogeneity is a significant issue within our sample. Nonetheless, as Hill et al. [141] emphasize, researchers and reviewers alike must acknowledge that no single study can eliminate all endogeneity-related concerns.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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